

SPOT EARLY

Team No Name · Défi 1.2 · dental screening at school · Reviewed by IEEE ISTMT Student Branch · 18 September 2026

Through to the pitch day | 78 points (75 core + +3 bonus) | Strong

HOW WE READ YOUR PROJECT

We scored you as a trained-model team working on public data: 2,211 anonymised intraoral images from Zenodo, fine-tuned into a YOLOv8n detector behind a React and FastAPI screening app. Detection rather than classification is the right call when a dentist needs to see where the lesion is, and saying no to hardware is correct when the camera is already the sensor. We installed your project and ran it ourselves, so what follows comes from your code running rather than from your description of it.

What you sent

GitHub repository	yes
Reading guide	yes and it worked
Pitch deck	yes 15 slides
Demo video	yes
Live demo (optional)	no
Hardware visual	no n/a
Data sources	yes Zenodo + 2 papers
Privacy and ethics	no left blank

Where the points went

A Working prototype	22/25	We followed your guide as written, both services came up, and /detect answered correctly on the first request.
B Technical design	13.5/15	Your architecture slide is exactly what your code does, and the deck explains why you fine-tuned rather than only that you did.
C Data and intelligence	13/15	The dataset audit and the separate held-out test split are the most careful data work in this round.
D Privacy and security	7/20	Children's names, an ID number and intraoral photos sit unencrypted in browser storage, with no accounts and no consent step.
E Process and honesty	9/10	Three concrete hard problems, one of them openly unsolved, and limitations that matched everything we found.
F User and local context	6.5/10	The five-angle capture protocol is real clinical thinking. The interface is English only, for a Tunisian school.
G Roadmap and feasibility	4/5	Clear items that each say what they add, but only one carries an estimate.
Bonus	+3	Known limitations, candid and accurate. Nothing you disclosed turned out to be understated.

WHAT WORKED

- **Your reading guide actually works.** pip install, uvicorn, npm install, npm run dev, no guessing anywhere. That is rarer than it should be.
- **You reported your model as worse than it is.** Your notebook's held-out test split gives precision 0.853, recall 0.910, mAP50 0.944, mAP50-95 0.743. Your form says "above 90%" and "roughly 75%". Claiming less than your own numbers show is the behaviour we most want to see, and reporting recall on a screening task is the right instinct.
- **The model stays quiet on a healthy mouth.** clear.jpeg returns zero detections while the others return three to five. We drew the boxes back onto the image and every one sat on a molar with visible occlusal staining. Not firing on everything is the harder half of screening.

WHAT HELD IT BACK

- **Your split may not separate patients.** It is stratified by image group and angle, not by case. Your filenames carry a case identifier (anonymous_003-007-673-00), so several views of the same mouth can land in training and testing at once. If that is happening, all four of your metrics are optimistic. We could not check without the dataset, so treat it as a question rather than a verdict.
- **The interface is English only.** The people who would run this are Tunisian school health staff working in Arabic and French, and there is no offline or low-end device story for a school that may have neither a good connection nor a machine to run the backend on.
- **npm run build fails.** PatientFile.tsx line 45 declares detections and never reads it, so the production build stops on TS6133 even though the dev server is fine.

SECURITY AND PRIVACY

Done right. Nothing leaks from the repo. No secrets committed, CORS pinned to localhost, and both the content type and the image itself are validated before the model sees them. Nothing you process leaves the machine it runs on.

Needs fixing.

- **You left the privacy question blank, and the code shows why that mattered.** Full name, age, an identity number and intraoral photographs of children go into browser localStorage in plain text.
- **No login, no roles.** /dashboard is open to anyone who reaches the app. A tool built around "professional validation" needs at least two roles to mean anything.
- **Nothing on consent.** Screening a minor in a school needs parental authorisation, and there is no mention of 2004-63, the INPDP, or how long the data stays.

CLAIMS WE COULD NOT MATCH TO THE EVIDENCE

- Your landing page says "Built and tested on real, consented screening images, never simulated data", while the app ships a clearly labelled path whose own text reads "This demo simulates the AI processing step". The labelling where it counts is honest, so we have not treated this as a real contradiction, but the two lines should not both be in the product.
- The confusion matrix figures in your form (919 / 258 / 72) are on slide 10 of your deck but not in the repo. Committing the ultralytics output would close that.

WHERE TO TAKE IT NEXT

- **Re-split by case, then rerun the evaluation.** Group on the case identifier in the filename before splitting, retrain, and publish whatever comes out. Half a day, and it turns your numbers into ones you can defend to a clinician.
- **Drop the identity field and add a consent tick.** Store a screening code instead of the child's name and ID, and put a short section in the README saying where the data sits and how long it stays.
- **Delete one line to fix the build,** then add Arabic and French strings. The UI text is small enough that this is an afternoon.